23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No : D-11
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Project Title : E-Commerce Platform

Abstract:

This project outlines the design and implementation of a full-stack e-commerce platform. With the rapid growth of online shopping, businesses require efficient and secure systems to manage customers, products, inventory, and orders while providing a seamless shopping experience. Traditional manual methods of handling these operations are time-consuming, prone to errors, and difficult to scale. This project aims to address these challenges by developing a reliable database-driven e-commerce platform that ensures efficient data management, secure transactions, and improved user experience.

Developed to mimic core functionalities of enterprise systems like Amazon, the application leverages a robust three-tier architecture to ensure scalability, data integrity, and a seamless user experience. The frontend is built using standard HTML5 and CSS3, providing a responsive and intuitive interface for product browsing, shopping cart management, and user authentication. The backend architecture is driven by Python, utilizing the lightweight Flask framework to handle server-side logic, routing, and HTTP request processing. Flask acts as the secure intermediary, processing user actions from the frontend and executing corresponding database operations.

The core of the system is powered by PostgreSQL, an advanced object-relational database management system (DBMS). The database schema enforces strict relational integrity across critical entities, including Users, Products, Categories, Orders, and Order Items. Complex SQL queries, transactions, and foreign key constraints ensure that inventory levels adjust dynamically upon purchase, order histories remain immutable, and user data is securely isolated.

The system is intended for customers and administrators. Customers can register, log in securely, browse products, search for items, add products to their shopping cart, place orders, and view their order history. Administrators can manage products, categories, inventory, customer information, and order details through the system, ensuring efficient and organized database management.

The proposed system is feasible to implement within the project duration and demonstrates advanced database concepts such as transaction management, relational integrity, inventory tracking, and secure user authentication, making it more comprehensive than a basic database management application.